SKY TECH ACADEMY
ROPE ACCESS TRAINING CENTER

OILFIELD | INDUSTRY | SAFETY | EXPERTISE

ROPE ACCESS TRAINING

SPRAT & IRATA Certification Programs

Houston, Texas | Industrial Rope Access Training Center

SPRAT	IRATA
Levels 1 · 2 · 3	Levels 1 · 2 · 3

Serving: Petrochemical | Oil & Gas | Construction | Wind Energy

MODULE 1 — Introduction to Rope Access

1.1 What Is Rope Access?

Rope access is an industrial work method that uses techniques adapted from caving and rock climbing to move up, down, or across ropes to reach elevated or difficult-to-access locations. It is widely recognized as one of the safest and most efficient methods for working at height.

At Sky Tech Academy, students will learn that rope access is not simply a climbing skill — it is a structured, regulated profession built on a foundation of safety, redundancy, and technical competence.

Key Principles of Rope Access

- Origin from caving and rock climbing techniques, adapted for industrial use
- Considered safer than many traditional access methods such as scaffolding
- Always uses two independent rope systems simultaneously:
  - Working rope (main line) — primary load-bearing system
  - Safety rope (backup line) — independent secondary protection
- All critical components are backed up — anchors, ropes, and attachment points
- Uses front (ventral) attachment points for better control and positioning
- Specialized lightweight equipment designed specifically for industrial operations
- Workers remain continuously supported — known as the 'post-fall' working position
- Fast and efficient — significantly reduces mobilization time vs. scaffolding

Key Highlight: *Industrial Rope Access is a safe, efficient, and highly controlled method of working at height using dual-rope systems and specialized training.*

1.2 Rope Access vs. Scaffolding

Comparison Factor	Rope Access	Scaffolding
Flexibility	Flexible and mobile — adapts to complex structures	Fixed structure — limited to planned access points
Setup Time	Quick setup — typically hours	Time-consuming erection and dismantling
Hazard Exposure	Lower exposure to hidden hazards	Users may not know scaffold-specific risks
Training	Requires specialized certification (SPRAT/IRATA)	Often used by general workers with minimal training

Comparison Factor	Rope Access	Scaffolding
Cost	Up to 62% cost reduction in documented studies	Significantly higher material and labor costs
Team Size	Small, highly trained teams	Larger workforce required

1.3 Safety Advantages of Rope Access

When a fall occurs using a conventional fall arrest system (such as scaffolding with dorsal D-ring attachment), the worker is suspended between the shoulder blades. This position can be uncomfortable, unstable, and may make self-rescue difficult. Prolonged suspension in this position can lead to suspension trauma (orthostatic intolerance), which may become life-threatening within approximately 20 minutes if rescue is not performed.

Rope access systems are fundamentally different. The dual-rope system means technicians are always connected to two independent lines, dramatically reducing the risk of a fall occurring at all. Key safety features include:

- Dual protection — working line and safety backup line at all times
- Reduced shock loading — workers are pre-supported, not falling
- Lower fall factor — backup device positioned as high as practical
- Controlled working position — stable, hands-free operation
- Smaller teams — fewer personnel exposed to risk

> **Key Highlight:** *Rope access safety is built on redundancy and controlled fall risk — dual independent systems prevent failure while fall factor management minimizes consequences of any potential fall.*

1.4 Cost Efficiency

Beyond safety, rope access delivers significant financial benefits. A study conducted at an industrial refinery comparing traditional scaffolding with rope access for the same scope of work found that rope access reduced overall project costs by approximately 62%. This savings comes from reduced material costs, smaller teams, faster mobilization, and lower overall exposure hours.

1.5 Rope Access as an Access Method — Not the Work Itself

A critical concept for all Sky Tech Academy students: rope access is a method of reaching the worksite — it is not the work itself. Technicians will receive additional training specific to their trade or assigned tasks, which may include:

- Inspection and Non-Destructive Testing (NDT)
- Maintenance and mechanical work
- Installation and construction tasks

- Painting, coating, and surface preparation
- Other specialized industrial services

> **Key Highlight:** *Rope access provides safe access — professional competence comes from combining rope access skills with your trade or technical discipline.*

MODULE 2 — Fall Protection Principles & Regulations

2.1 Fall Protection Overview

Fall protection refers to the systems and measures used to keep workers safe while working at height, either by preventing a fall from occurring or safely arresting a fall if it does occur. Fall protection programs are established and enforced by the employer, who is responsible for ensuring a safe working environment.

2.2 The Three Types of Fall Protection Systems

TYPE 1 — Fall Arrest

Designed to stop a worker after a fall has already occurred. Typically involves a full body harness, lanyard or SRL, and anchorage point. Fall arrest is considered less effective than prevention because some force is still transferred to the body.

TYPE 2 — Fall Restraint / Travel Restraint

Prevents the worker from reaching a fall hazard in the first place by restricting movement. A properly adjusted lanyard on a horizontal lifeline, for example, allows movement within a safe zone but prevents reaching the edge. This method effectively eliminates fall risk.

TYPE 3 — Work Positioning

Supports part or all of the worker's weight through ropes and associated equipment. Widely used in rope access, this system allows the technician to maintain a stable, hands-free working position on structures such as walls, towers, and vertical surfaces.

2.3 Fall Factor

Fall Factor indicates the severity of a potential fall. It is calculated by dividing the maximum distance a worker could fall by the total length of the rope or connection between the worker and the anchor point. A higher fall factor represents a more severe fall and increased stress on the system.

Backup Position	Fall Distance	Rope Length	Fall Factor
Backup high (as practical)	0 ft	3 ft	FF = 0 (Ideal)
Backup at attachment level	3 ft	3 ft	FF = 1
Backup very low (by feet)	6 ft	3 ft	FF = 2 (Maximum risk)

Key Highlight: *In rope access, proper positioning of the backup device at or above head level minimizes fall factor and reduces forces on the body.*

2.4 Harness Attachment Points

Each attachment point on a rope access harness serves a specific function and must be used correctly. Misuse of attachment points is a leading cause of preventable incidents.

Attachment Point	Location	Primary Use	Rated Load	Key Notes
Ventral	Front waist	Rope access / work positioning	~15 kN	Primary load-bearing point for rope access
Sternal	Chest	Backup / chest ascender (Croll)	~15 kN	Used with chest ascenders and redundancy
Dorsal	Back	Fall arrest ONLY	8 kN peak / 4 kN avg	NOT used for rope access positioning
Lateral	Side D-rings	Work positioning	~15 kN	ALWAYS used in pairs — never individually

Key Highlight: *The ventral point is the primary support for rope access — the dorsal point is reserved exclusively for fall arrest systems.*

2.5 Force Limits by Attachment Point

Connection	Peak Arrest Force	Average Force	Max Free-Fall Distance
Sternal (front)	900 lb (4 kN)	450 lb (2 kN)	2 feet
Dorsal (back)	1,800 lb (8 kN)	900 lb (4 kN)	6 feet (unless otherwise certified)

2.6 Anchorage Requirements

- Non-engineered anchorages (fall arrest / work positioning): minimum 5,000 lb (22.2 kN) per user
- Engineered anchorages: typically 3,600 lb (16 kN) — must support twice maximum expected load
- Travel restraint systems: minimum 1,000 lb (4.45 kN)
- All D-rings: minimum breaking strength (MBS) of 5,000 lb (22.2 kN)
- Harness buckles: minimum MBS of 4,000 lb (17.8 kN)

2.7 Regulatory Framework — OSHA & ANSI

In the United States, fall protection is governed by OSHA (Occupational Safety and Health Administration) under two primary standards:

- 29 CFR 1910 — General Industry
- 29 CFR 1926 — Construction

ANSI (American National Standards Institute) develops voluntary performance and testing standards for fall protection equipment. However, when OSHA adopts ANSI standards into its regulations, compliance becomes mandatory.

Key ANSI Z359 standards relevant to rope access:

ANSI Standard	Title	Relevance to Rope Access
Z359.1	The Fall Protection Code	Minimum requirements for managed fall protection programs
Z359.2	Managed Fall Protection Program	Employer responsibilities and worker training requirements
Z359.4	Assisted Rescue and Self-Rescue Systems	Relevant to descenders and rescue devices used in rope access
Z359.11	Full Body Harnesses	Design and testing of harnesses (59–140 kg / 130–310 lbs)
Z359.14	Self-Retracting Lifelines (SRLs)	Requirements for SRL devices used alongside rope access systems

Key Highlight: *OSHA requires employers to implement structured fall protection programs aligned with ANSI standards, ensuring workers are protected through compliant equipment, procedures, and safe practices.*

2.8 Self-Retracting Lifelines (SRLs)

SRLs are designed to rapidly arrest a fall while minimizing free-fall distance. They use an internal inertial braking mechanism that activates when sudden acceleration is detected. Key considerations when using SRLs:

- Choose the correct attachment point (sternal or dorsal) based on device specifications
- Dorsal connection may result in forward-leaning fall — risk of impact with adjacent structures such as ladders
- Sternal attachment must be confirmed compatible with device force ratings
- Twin SRLs provide 100% tie-off but add weight and bulk
- Always inspect before use — verify force ratings and compatibility with work conditions

Key Highlight: *Safe SRL use depends on proper selection, correct attachment, and thorough understanding of how the device behaves in real fall scenarios.*

MODULE 3 — SPRAT Standards & Certification

3.1 What Is SPRAT?

The Society of Professional Rope Access Technicians (SPRAT) is the primary governing body for rope access work in North America. SPRAT provides a structured framework for training, certification, and safe work practices, and its certifications are widely required by industrial employers across the United States.

Website: www.sprat.org

3.2 SPRAT Key Documents

Document	Purpose
Safe Practices for Rope Access Work	Defines accepted methods and procedures for performing rope access tasks safely
Certification Requirements for Rope Access Work	Establishes criteria for training and certifying rope access technicians
Evaluation Guidelines	Insight into how certification assessments are conducted and what is expected
Evaluation Matrix	Outlines potential errors and how performance is graded during assessment
Defined Terms	Ensures consistent understanding of key terminology across all SPRAT documents

3.3 SPRAT Certification Levels

LEVEL 1 — Rope Access Technician — Entry Level

- Entry-level worker; requires the most supervision
- No minimum experience or work-hour requirements — intended for new entrants
- Must complete training and sign SPRAT training affidavit
- Works under direct supervision of Level 2 or Level 3

LEVEL 2 — Rope Access Technician — Intermediate

- Minimum 500 hours of rope access work experience
- Minimum 6 months field experience
- Can provide limited supervision to Level 1 under Level 3 direction
- Capable of advanced rigging including deviations
- Can perform more complex rescue operations

LEVEL 3 — Rope Access Supervisor — Advanced

- Minimum 1,000 hours of rope access work experience
- Minimum 12 months field experience
- Acts as Rope Access Supervisor — full accountability for site safety
- Responsible for pre-job safety meetings, JSA/JHA, hazard control
- Plans and leads emergency rescue operations
- Maintains worker experience records

3.4 SPRAT Certification Requirements

To become SPRAT certified, candidates must meet the following basic requirements:

- Minimum age of 18
- Completion of a liability waiver and declaration of physical and mental fitness
- Demonstrated aptitude and appropriate attitude for working at height
- Completion of a SPRAT training affidavit confirming adequate training

SPRAT does not specify a fixed number of training hours. Instead it focuses on demonstrated competence. In practice, Sky Tech Academy recommends:

Level	Recommended Training Hours	Notes
Level 1	Minimum 40 hours (including evaluation)	Mandatory initial training
Level 2	Minimum 40 hours (including evaluation)	Training required for first attempt at each level
Level 3	120+ hours	Reflects increased supervisory responsibility

3.5 SPRAT Assessment & Grading

SPRAT certification is performance-based. Each skill is evaluated using three possible ratings:

Rating	Code	Description
Pass	P	Task completed safely and correctly
Discrepancy	D	Minor safety or procedural error (e.g., unlocked carabiner, unsecured helmet strap)
Fail	F	Task not completed safely

Automatic failure conditions:

- A single Fail (F) in any task = overall failure
- Three Discrepancies (D) = overall failure

- Relying on a single rope instead of maintaining redundancy = immediate failure
- Improper or ineffective use of backup devices = immediate failure
- Inability to complete an exercise without assistance or rescue = immediate failure

Key Highlight: *SPRAT certification follows a structured three-level system with strict assessment criteria, ensuring technicians demonstrate both technical competence and a strong commitment to safety.*

MODULE 4 — IRATA Standards & Certification

4.1 What Is IRATA?

The Industrial Rope Access Trade Association (IRATA) is the internationally recognized standard for rope access, widely used in offshore oil & gas, construction, and energy sectors globally. IRATA has been established longer than SPRAT, reflecting the earlier adoption of rope access techniques in Europe. While both organizations share similar safety goals, IRATA documentation is significantly more detailed and comprehensive.

4.2 International Code of Practice (ICOP)

IRATA's primary document is the International Code of Practice (ICOP), divided into four parts:

ICOP Part	Title	Content Summary
Part 1	General Information	Introduction, scope, terminology, and basic principles of rope access. Defines key elements of a safe system of work.
Part 2	Detailed Guidance	In-depth guidance on planning, personnel selection, equipment requirements, and practical work methods from start to finish.
Part 3	Informative Annexes	Risk assessment, equipment inspection, anchor systems, suspension trauma, rope access techniques, and environmental factors.
Part 4	Local Legislation	Region-specific legal requirements (primarily UK). Provides useful regulatory insight.

4.3 IRATA Key Safety Principles (ICOP Part 1)

- Proper planning and effective management of all operations
- Use of trained and competent personnel at all times
- Adequate supervision — only a Level 3 can act as safety supervisor
- Careful selection of suitable equipment for the task
- Proper inspection, maintenance, and traceability of all equipment
- Maximum impact force on a falling technician limited to 6 kN (~1,348 lbf)
- Mandatory exclusion zones — anchor area, working edge, intermediate level, and bottom level
- Documented and communicated rescue procedures before work begins
- Continuous improvement through internal and IRATA audits

Key Highlight: *IRATA limits maximum impact force to 6 kN — stricter than OSHA's 8 kN allowance — reflecting its focus on controlled systems and fall prevention over arrest.*

4.4 IRATA Certification Levels (TACS)

The IRATA Training, Assessment, and Certification Scheme (TACS) defines how technicians are trained, assessed, and certified. It is mandatory for all IRATA member companies.

LEVEL 1 — Operative — Entry Level

- Entry-level; requires close supervision from Level 3 at all times
- No minimum experience requirement
- Must follow procedures, method statements, and risk assessments
- Perform pre-use equipment checks
- Assist in rigging under supervision
- Carry out simple rescues and assist in complex ones

LEVEL 2 — Technician — Intermediate

- Minimum 1,000 hours of rope access work experience
- Minimum 12 months field experience
- Must demonstrate all Level 1 skills
- Perform advanced rigging: deviations, re-belays, and tensioned lines
- Carry out rescues in a variety of scenarios
- Assemble and operate hauling systems

LEVEL 3 — Supervisor — Advanced

- Minimum 2,000 hours of rope access work experience
- Minimum 24 months field experience
- Responsible for supervision and overall site safety
- Strong knowledge of IRATA's safe system of work and regulations
- Highly skilled in advanced rigging and all rescue scenarios
- Must hold a valid first aid certificate

4.5 IRATA Assessment Process

Key differences from SPRAT that candidates should note:

- Formal training with an approved IRATA provider is mandatory before assessment
- Assessment must occur within 60 days of completing training
- Current certification must be valid on the day of assessment — expired certs may prevent assessment
- Direct entry pathway available for candidates with significant prior experience
- Grading: Pass / Discrepancy / Fail — three discrepancies or any major error = failed assessment
- Re-assessment typically permitted within 60 days of failure

- Logbooks must be properly maintained to track all work experience hours

Key Highlight: *IRATA requires structured training, strict assessment timelines, and strong personal responsibility — ensuring only competent, prepared, and physically fit technicians are certified.*

4.6 SPRAT vs. IRATA — Comparison at a Glance

Factor	SPRAT	IRATA
Primary Region	North America (USA)	International — offshore, Europe, global
Documentation Length	~50 pages (concise)	Several hundred pages (comprehensive)
Level 2 Experience Req.	500 hrs + 6 months	1,000 hrs + 12 months
Level 3 Experience Req.	1,000 hrs + 12 months	2,000 hrs + 24 months
Max Impact Force	Per OSHA: 8 kN dorsal / 4 kN sternal	6 kN (stricter limit)
Training Hours	No fixed requirement (40 hrs recommended)	Formal approved training mandatory
Assessment Window	No fixed requirement	Within 60 days of training
Audit Requirement	Company-level	Internal + formal IRATA audits
First Aid Requirement	Not specified at L1/L2	Mandatory for Level 3

MODULE 5 — Roles & Responsibilities

5.1 Employer Responsibilities

Under both SPRAT and IRATA, the employer bears primary responsibility for establishing and maintaining a safe rope access program. This includes:

- Developing a clear rope access policy with safety goals, procedures, and expectations
- Providing necessary resources, equipment, and oversight
- Appointing a Rope Access Program Administrator (required by SPRAT)
- Ensuring written rope access and rescue procedures are available at every worksite
- Staffing each worksite with qualified Level 3 supervision
- Verifying that each worker's training and experience are valid before assignment
- Including hazard identification tools such as JSA/JHA in site procedures

Key Highlight: *Employers play a critical role in rope access safety by establishing structured programs, providing resources, ensuring proper supervision, and verifying worker competence before any work begins.*

5.2 Rope Access Program Administrator

This role is specifically required by SPRAT and is typically held by a Level 3-equivalent individual. The Program Administrator is responsible for:

- Developing and implementing the rope access program
- Monitoring effectiveness and ensuring regulatory compliance
- Reviewing and updating procedures as standards change
- Selecting qualified personnel for rope access tasks
- Maintaining worker training, experience, and equipment records
- Leading incident investigations and implementing corrective actions
- Serving as the main client contact prior to mobilization
- Prioritizing safety over productivity or financial pressures at all times

5.3 Level 1 Technician — Duties & Responsibilities

A Level 1 Rope Access Technician is an entry-level worker who must operate under supervision of a Level 2 or Level 3 at all times.

- Demonstrate basic competence in rope access operations
- Perform simple rescue tasks and assist higher-level technicians in complex rescues
- Know and follow the employer's rope access program policies and procedures
- Properly inspect, use, maintain, and store personal rope access equipment
- Check integrity of anchor systems, slings, and connectors
- Identify and report all hazards to the Level 3 supervisor immediately

- Participate in development and review of safety documentation (JSA/JHA, permits)
- Understand and operate appropriately within access zones, safe zones, and hazard zones
- Use all required PPE including respiratory protection when specified
- Follow all supervisor instructions — ask for clarification if unclear
- Inform supervision if a task is beyond training or experience
- Stop work if a task cannot be performed safely or immediate danger exists

5.4 Level 2 Technician — Duties & Responsibilities

A Level 2 technician builds on all Level 1 skills with additional capabilities in supervision, rigging, and rescue.

- Perform all Level 1 duties
- Provide limited supervision to Level 1 technicians — only under Level 3 direction and approval
- Cannot assume supervisory role independently; Level 3 must formally designate this
- Ensure proper use of PPE and identify and mitigate hazards
- Stop work if unsafe conditions arise
- Build, assess, and modify rope access systems including deviations
- Perform complex rescue operations
- Assist with updates to safety documents including permits and JSAs

Key Highlight: *Level 2 technicians bridge the gap between worker and supervisor — combining advanced technical skills with limited leadership responsibilities under Level 3 direction.*

5.5 Level 3 Technician — Duties & Responsibilities

The Level 3 acts as the Rope Access Supervisor and bears full accountability for the safety, efficiency, and success of all rope access operations.

- Full control, oversight, and accountability for the worksite and all personnel
- Strong understanding of all relevant regulations, company procedures, and client requirements
- Identify hazards through risk assessments, JSA/JHA, and implement appropriate controls
- Ensure no work begins without appropriate hazard controls in place
- Conduct pre-job safety meetings and review all required safety documents
- Select and enforce proper use of all PPE
- Establish and control work zones — only authorized personnel in hazardous areas
- Plan and lead emergency response including rescue procedures and coordination with emergency services
- Ensure all rope access equipment is selected, inspected, and properly maintained
- Remove damaged or unsafe equipment from service immediately
- Maintain accurate records of work hours and experience for each technician
- Remain competent in all Level 1 and Level 2 skills

Key Highlight: *The Level 3 technician is the leader of the rope access operation — responsible for safety, supervision, planning, equipment, and emergency response, with full accountability for the entire worksite.*

MODULE 6 — Training Guidelines at Sky Tech Academy

6.1 Sky Tech Academy Training Philosophy

At Sky Tech Academy, every training program is designed around one guiding principle: safety through competence. We prepare technicians not just to pass their certification assessments, but to perform confidently and safely in real industrial environments across Houston's petrochemical, oil & gas, construction, and wind energy sectors.

6.2 Course Structure Overview

Course	Duration	Prerequisite
SPRAT Level 1	5 days	Age 18+, physical fitness declaration
SPRAT Level 2	5 days	SPRAT L1 + 500 hrs + 6 months field
SPRAT Level 3	5 days	SPRAT L2 + 1,000 hrs + 12 months field
IRATA Level 1	5 days	Age 18+, physical fitness declaration
IRATA Level 2	5 days	IRATA L1 + 1,000 hrs + 12 months field
IRATA Level 3	5 days	IRATA L2 + 2,000 hrs + 24 months field
SPRAT/IRATA Dual Level 1	5 days	Age 18+, physical fitness declaration
Refresher Courses	2 days	Current or recently lapsed certification
Advanced Rescue	2 days	Level 3 certification

6.3 Training Rules & Safety Requirements

All Sky Tech Academy students are required to observe the following rules throughout training:

- Strictly follow all instructions given by the instructor at all times
- Never rely on a single rope system — always maintain dual protection
- Never perform rope access activities alone — always work with a qualified partner
- Report any uncertainty or hazard to the instructor immediately
- Perform warm-up and stretching exercises before every practical session
- Failure to follow instructions may result in removal from the training program

Rope access training requires participants to be in good physical and mental condition. It is the responsibility of each student to inform the instructor of any medical condition that may affect their ability to safely participate in training activities.

Key Highlight: *Rope access requires ongoing training, strict adherence to supervision, and a clear understanding that it is an access method — not the work itself. Safety is always the highest priority at Sky Tech Academy.*

6.4 Continuing Professional Development

- Recertification is required every 3 years for SPRAT and annually for IRATA
- Regular refresher training is strongly recommended between certification cycles
- Level 3 technicians are encouraged to attend advanced rescue and rigging workshops
- Sky Tech Academy offers corporate on-site training programs for team certification
- All training records and logbooks must be maintained accurately by each technician

6.5 Company Procedures & Compliance

In addition to SPRAT and IRATA standards, every company that operates a rope access program must establish its own procedures. These typically include:

- Safe working practices specific to the company's operations and industry sector
- Standard rescue procedures tailored to the work environment
- Hazard identification and risk control processes (JSA/JHA)
- Equipment inspection, use, maintenance, and disposal protocols
- Company-specific rules and operational requirements

All Sky Tech Academy graduates are expected to read, understand, and strictly follow the rope access procedures of their employer. Lack of knowledge is not considered an acceptable excuse for non-compliance.

Key Highlight: *A successful rope access program depends on structured training, clear company procedures, and strong coordination between all elements — ensuring safe, consistent, and efficient operations.*

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